Pengyu Zhang

PhD Candidate in Computer Science · University of Amsterdam

Amsterdam, the Netherlands · p.y.zhang3@gmail.com · pengyu-zhang.github.io

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About Me

I work on machine learning for knowledge graphs, where the data keeps changing and many entities look alike. I build models that add signals which are usually left out, such as time and how the graph changes, so that entity linking, link prediction and recommendation stay accurate on sparse, long-tail and look-alike cases. The same problem appears in industry as entity resolution under data drift.

Research interests: multi-modal knowledge graphs, entity linking and resolution, link prediction, recommendation, large language models.

Education

2022 – Present **PhD in Computer Science** (defense planned for March 2027), INDElab, Faculty of Science, University of Amsterdam (UvA), the Netherlands. Supervisors: Prof. Dr. Paul Groth, Dr. Klim Zaporojets.
Thesis: *Addressing Temporal Dynamics and Entity Ambiguity in Knowledge Graph Tasks with Multi-Source Information*.

2019 – 2022 **MEng in Control Engineering**, Faculty of Information Technology, Beijing University of Technology (BJUT), China. Supervisor: Prof. Dr. Yong Zhang.

Experience

PhD Candidate, INDElab, University of Amsterdam 2022 – Present
Five first-author and five co-authored papers. Every published first-author paper has public code.

TimeRoute Time-aware multi-modal recommendation Under review

- Which modality matters for a purchase changes over time and differs from user to user, but most recommenders fuse text, images and audio with one global weighting. I route the modalities for each user based on their own interaction times, and use time-conditioned diffusion to filter out outdated edges.
- On three datasets this beats the strongest baseline on every metric, by up to 9.8%. On one of them I gave the router noise instead of time, and its gain disappeared.

Time Imprint Time as a modality for look-alike entities Under review

- Some entities have almost the same text and images, like Napoleon Bonaparte and the 2023 film *Napoleon*, but their years are very different. I treat time as a modality of each entity: its own timestamps are pooled into one signal and used in the encoder, the scoring function and a contrastive objective.
- Link prediction improves most on the one percent of entities that are hardest to tell apart, where Hits@1 rises by up to 58.2%.

Select, Don't Train LLM-based entity disambiguation ISWC 2026

- With Fina Polat (lead), we split entity disambiguation into two steps instead of one dual encoder: a plain BM25 retriever finds the candidates, and an LLM picks one or says that none of them fits.
- Without training any retriever, this beats the best trained dual encoder on the ZELDA benchmark by 4 points of micro-F1.

Fusion Training Hybrid-reasoning LLMs ACL 2026 SRW

- Hybrid models like Qwen3 can answer quickly or think step by step with the same weights. With Congfeng Cao (lead), we varied how the two kinds of training data are mixed and scheduled on Qwen3-4B.
- More short-answer data lowers the accuracy of long-form reasoning, and interleaving the two kinds is the most robust schedule.

Beyond Images Knowledge-graph data enrichment ESWC 2026

- Curated image sets leave out ambiguous pictures such as logos and symbols, because visual encoders turn them into noise. I collect all Wikipedia images of an entity (more than 300,000 new images in total), caption each one with a vision-language model and let an LLM fuse the captions into one text summary.

- Without any change to the models, link prediction improves on all three datasets and four models I tested, by up to 7.8% in Hits@1, and most on entities whose images are mainly logos or symbols.

Graph-TempCZ Large-scale temporal link prediction LREC 2026

- With Congfeng Cao (lead), we turned software mentions from 1.5 million biomedical papers into a publication-software graph with a time axis, and framed software usage as temporal link prediction.
- A GraphSAGE model does better than feature-based baselines, and its accuracy drops as the gap between the training year and the test year grows.

Graph Learning Challenges LLMs for data challenges in graphs ESWA 2026

- A survey. With Mengran Li (lead), we organised more than 380 papers around four data challenges of real graphs: missing parts, imbalance, different domains and change over time.
- My part was the section on graphs that change over time.

Understanding Entity Linking Entity linking for social media analysis EKAW 2024

- With James Nevin (lead), we ran two established entity linkers at several thresholds over 21 million tweets, and the linking pipeline was my part.
- The two linkers agree on fewer than half of the entities, and the networks built from them differ in shape and even in their top entities.

CYCLE & TIGER Temporally robust entity linking CIKM & ECAI 2024

- An entity linker is trained on one snapshot of the knowledge graph and then used for years while the graph keeps changing, so its accuracy decays. In TIGER I add graph structure, so entities can still be told apart when their descriptions drift. In CYCLE I use the changes as supervision: relations that appear or disappear between snapshots become contrastive examples.
- Both beat the 2023 state of the art on the public yearly-snapshot benchmarks I built for them. At a three-year gap, TIGER is 20.9% ahead on continual entities and CYCLE 17.8% ahead averaged over continual and new entities.

MEng Student, Beijing University of Technology, group of Prof. Dr. Yong Zhang 2019 – 2022
Graph learning and author name disambiguation.

RUNCL Graph structure learning for low-degree nodes Physica A 2024

- GNNs take the input graph as correct, so missing or wrong edges hurt the sparsely connected nodes most. I generate candidate neighbourhood graphs from node features, choose the most likely one, and let the feature view and the relationship view supervise each other through contrastive learning.
- On six benchmarks it is best in most settings with few labels.

MVMA-GCN Multi-view graph learning EAAI 2023

- Real networks have several kinds of links, such as co-author and co-keyword, but most GNNs use one view at a time or give all views the same weight. I feed all views into the model together: two attention layers decide how much each neighbour and each view counts, and an HSIC term keeps the views distinct.
- On eight benchmarks it beats the strongest baseline in most settings.

Visual Analysis Interactive author name disambiguation JCAD 2022

- Research offices spend a lot of manual effort separating authors who share a name, and a classifier alone gives them no way to check its decisions. I built and deployed a system that pairs a multi-view graph classifier with linked views of the collaboration network and the raw metadata, so users can still decide when the classifier is wrong.
- In a user study on 4,000 real university papers, graduate students and experienced administrators completed the tasks with more than 95% accuracy.

DualChannel Author name disambiguation at scale Information 2021

- With Xin Zheng (lead), we combined the text of each paper with a graph of co-authors, organisations and venues, then clustered the papers so that the number of authors does not have to be set in advance.
- On AMiner WhoIsWho, about 205,000 papers, it beats all the baselines we compared.

Awards & Service

AWARDS	Conference travel grant for ACL 2026. Outstanding Master's Thesis, BJUT, 2022.
PRESENTATIONS	Papers at ACL SRW, ESWC and LREC 2026, and at EKAW, CIKM and ECAI 2024.
SERVICE	Reviewer / sub-reviewer: CIKM, ECAI, ESWC, ECML PKDD.
TEACHING	Teaching assistant, Data Engineering (BJUT), 2020 – 2021.

Skills

PROGRAMMING	Python, PostgreSQL, Bash, Git, Linux
ML TOOLS	PyTorch, Hugging Face, scikit-learn, NumPy, pandas, Weights & Biases; multi-GPU training on A100 and H100
METHODS	Large language models, Vision-language models, Graph neural networks, Contrastive learning, Entity linking and resolution, Recommendation, Multi-modal fusion, Diffusion models, Temporal representation learning
LANGUAGES	Chinese (native), English (fluent, IELTS 7.0), Dutch (elementary, A2)

Publications

[14] **TimeRoute: Time-Aware Modality Routing and Diffusion for Multi-Modal Recommendation. Pengyu Zhang**, Yangqin Jiang, Klim Zaporojets, Congfeng Cao, Paul Groth. Under review **arXiv**

[13] **Time Imprint: Learning Time-Aware Representations in Multi-Modal Knowledge Graphs. Pengyu Zhang**, Klim Zaporojets, Congfeng Cao, Jia-Hong Huang, Paul Groth. Under review **arXiv**

[12] **Select, Don't Train: The Benefits of Modular Entity Disambiguation with LLM-Based Selection.** Fina Polat, Daniel Daza, **Pengyu Zhang**, Klim Zaporojets, Paul Groth. ISWC 2026 **arXiv** **DOI** **Code**

[11] **Fusion Training for Mathematical Generalization in Large Language Models.** Congfeng Cao, **Pengyu Zhang**, Jelke Bloem. ACL 2026 SRW **Paper** **DOI** **Code**

[10] **Are a Thousand Words Better Than a Single Picture? Beyond Images - A Framework for Multi-Modal Knowledge Graph Dataset Enrichment. Pengyu Zhang**, Klim Zaporojets, Jie Liu, Jia-Hong Huang, Paul Groth. ESWC 2026 **Paper** **DOI** **Code** **Demo** **YouTube** **Bilibili**

[9] **Graph-TempCZ: A Graph Representation of Software Mentions for Predicting Software Usage in Scientific Publications.** Congfeng Cao, **Pengyu Zhang**, Jelke Bloem. LREC 2026 **Paper** **DOI** **Code**

[8] **A Survey of Large Language Models for Data Challenges in Graphs.** Mengran Li, **Pengyu Zhang**, Wenbin Xing, Yijia Zheng, Klim Zaporojets, Junzhou Chen, Ronghui Zhang, Yong Zhang, Siyuan Gong, Jia Hu, Xiaolei Ma, Zhiyuan Liu, Paul Groth, Marcel Worring. Expert Systems with Applications. ESWA 2026 **Paper** **DOI** **Code**

[7] **Understanding the Impact of Entity Linking on the Topology of Entity Co-occurrence Networks for Social Media Analysis.** James Nevin, **Pengyu Zhang**, Dimitar Dimitrov, Michael Lees, Paul Groth, Stefan Dietze. EKAW 2024 **Paper** **DOI** **Code**

[6] **CYCLE: Cross-Year Contrastive Learning in Entity-Linking. Pengyu Zhang**, Congfeng Cao, Klim Zaporojets, Paul Groth. CIKM 2024 **Paper** **DOI** **Code**

[5] **TIGER: Temporally Improved Graph Entity Linker. Pengyu Zhang**, Congfeng Cao, Paul Groth. ECAI 2024 **Paper** **DOI** **Code**

[4] **Relationship Updating Network with Contrastive Learning. Pengyu Zhang**, Yong Zhang, Xinglin Piao, Yongliang Sun, Baocai Yin. Physica A: Statistical Mechanics and its Applications. Physica A 2024 **Paper** **DOI** **Code**

[3] **MVMA-GCN: Multi-view Multi-layer Attention Graph Convolutional Networks. Pengyu Zhang**, Yong Zhang, Jingcheng Wang, Baocai Yin. Engineering Applications of Artificial Intelligence. EAAI 2023 **Paper** **DOI** **Code**

[2] **Visual Analysis for Name Disambiguation of Academic Papers *(in Chinese)*. Pengyu Zhang**, Yong Zhang, Yanjie Cui, Baocai Yin. Journal of Computer-Aided Design and Computer Graphics. JCAD 2022 **Paper** **DOI** **Code** **Demo** **YouTube** **Bilibili**

[1] **Dual-Channel Heterogeneous Graph Network for Author Name Disambiguation.** Xin Zheng, **Pengyu Zhang**, Yanjie Cui, Rong Du, Yong Zhang. Information 2021 **Paper** **DOI** **Code**